

Horseshoe_Crabs

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```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.4.4      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.0
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(fastDummies)
```

```
## Thank you for using fastDummies!
## To acknowledge our work, please cite the package:
## Kaplan, J. & Schlegel, B. (2023). fastDummies: Fast Creation of Dummy (Binary) Columns and Rows from
```

This problem refers to data from a study of nesting horseshoe crabs.

Each female horseshoe crab in the study had a male crab attached to her in her nest. The study investigated factors that affect whether the female crab had any other males, called satellites, residing near her. Explanatory variables that are thought to affect this included the female crab's color (1 = lightest in color to 5 = darkest in color), spine condition, carapace width (in centimeters), and weight. The response outcome for each female crab is the number of satellites. There are 173 females in this study.

```
Crab <- read_csv("Horseshoe_Crabs.csv") %>% as_tibble()
```

```
## Rows: 173 Columns: 6
## -- Column specification -----
## Delimiter: ","
## dbl (6): CRAB, COLOR, SPINE, WIDTH, SATELLITES, WEIGHT
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
summary(Crab)
```

```
##      CRAB      COLOR      SPINE      WIDTH      SATELLITES
## Min.   : 1   Min.   :2.000   Min.   :1.000   Min.   :21.0   Min.   : 0.000
## 1st Qu.: 44   1st Qu.:3.000   1st Qu.:2.000   1st Qu.:24.9   1st Qu.: 0.000
## Median : 87   Median :3.000   Median :3.000   Median :26.1   Median : 2.000
## Mean   : 87   Mean   :3.439   Mean   :2.486   Mean   :26.3   Mean   : 2.919
## 3rd Qu.:130   3rd Qu.:4.000   3rd Qu.:3.000   3rd Qu.:27.7   3rd Qu.: 5.000
## Max.   :173   Max.   :5.000   Max.   :3.000   Max.   :33.5   Max.   :15.000
##      WEIGHT
## Min.   :1.200
## 1st Qu.:2.000
## Median :2.350
## Mean   :2.437
## 3rd Qu.:2.850
## Max.   :5.200
```

```
Crab
```

```
## # A tibble: 173 x 6
##      CRAB COLOR SPINE WIDTH SATELLITES WEIGHT
##      <dbl> <dbl> <dbl> <dbl>      <dbl> <dbl>
## 1      1     3     3  28.3         8   3.05
## 2      2     4     3  22.5         0   1.55
## 3      3     2     1   26         9   2.3
## 4      4     4     3  24.8         0   2.1
## 5      5     4     3   26         4   2.6
## 6      6     3     3  23.8         0   2.1
## 7      7     2     1  26.5         0  2.35
## 8      8     4     2  24.7         0   1.9
## 9      9     3     1  23.7         0  1.95
## 10    10     4     3  25.6         0   2.15
## # i 163 more rows
```

```
Crab$COLOR <- as.factor(Crab$COLOR)
Crab
```

```
## # A tibble: 173 x 6
##      CRAB COLOR SPINE WIDTH SATELLITES WEIGHT
##      <dbl> <fct> <dbl> <dbl>      <dbl> <dbl>
## 1      1  1 3     3  28.3         8   3.05
## 2      2  2 4     3  22.5         0   1.55
## 3      3  3 2     1   26         9   2.3
## 4      4  4 4     3  24.8         0   2.1
## 5      5  5 4     3   26         4   2.6
## 6      6  6 3     3  23.8         0   2.1
## 7      7  7 2     1  26.5         0  2.35
## 8      8  8 4     2  24.7         0   1.9
## 9      9  9 3     1  23.7         0  1.95
## 10    10 10 4     3  25.6         0   2.15
## # i 163 more rows
```

```
Crab <- dummy_cols(Crab, select_columns = "COLOR", remove_first_dummy = TRUE)
Crab
```

```
## # A tibble: 173 x 9
##   CRAB COLOR SPINE WIDTH SATELLITES WEIGHT COLOR_3 COLOR_4 COLOR_5
##   <dbl> <fct> <dbl> <dbl>      <dbl>  <dbl>   <int>   <int>   <int>
## 1     1  1 3      3 28.3         8   3.05     1     0     0
## 2     2  2 4      3 22.5         0   1.55     0     1     0
## 3     3  3 2      1 26          9   2.3      0     0     0
## 4     4  4 4      3 24.8         0   2.1      0     1     0
## 5     5  5 4      3 26          4   2.6      0     1     0
## 6     6  6 3      3 23.8         0   2.1      1     0     0
## 7     7  7 2      1 26.5         0  2.35     0     0     0
## 8     8  8 4      2 24.7         0   1.9      0     1     0
## 9     9  9 3      1 23.7         0  1.95     1     0     0
## 10    10 10 4      3 25.6         0  2.15     0     1     0
## # i 163 more rows
```

(A) Fit a Poisson regression model to predict the number satellites for a female crab based on the predictor variables WIDTH (carapace width) and COLOR (female crab's color (2 – 5)). (4 points)

```
model11 <- glm(SATELLITES ~ WIDTH + COLOR_3 + COLOR_4 + COLOR_5, family = "poisson", data = Crab)
summary(model11)
```

```
##
## Call:
## glm(formula = SATELLITES ~ WIDTH + COLOR_3 + COLOR_4 + COLOR_5,
##     family = "poisson", data = Crab)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -2.65004    0.58802  -4.507 6.58e-06 ***
## WIDTH        0.14934    0.02084   7.166 7.73e-13 ***
## COLOR_3      -0.19969    0.15364  -1.300  0.1937
## COLOR_4      -0.43636    0.17636  -2.474  0.0133 *
## COLOR_5      -0.44736    0.20912  -2.139  0.0324 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 632.79  on 172  degrees of freedom
## Residual deviance: 559.34  on 168  degrees of freedom
## AIC: 924.64
##
## Number of Fisher Scoring iterations: 6
```

```
model11

##
## Call:  glm(formula = SATELLITES ~ WIDTH + COLOR_3 + COLOR_4 + COLOR_5,
##     family = "poisson", data = Crab)
##
## Coefficients:
## (Intercept)      WIDTH      COLOR_3      COLOR_4      COLOR_5
##    -2.6500      0.1493     -0.1997     -0.4364     -0.4474
```

```
##
## Degrees of Freedom: 172 Total (i.e. Null); 168 Residual
## Null Deviance: 632.8
## Residual Deviance: 559.3 AIC: 924.6
```

$$\widehat{\text{Satellites}}(Y = 1) = -2.6500 + 0.1493(\text{Width}) - 0.1997(\text{Color 3}) - 0.4364(\text{Color 4}) - 0.4474(\text{Color 5})$$

(B) Interpret the beta coefficient of WIDTH in the context of this problem. (5 points)

The beta coefficient of width in the Poisson regression model is 0.1493, and represents the coefficient of width with respect to the log of satellites. Thus we can exponentiate both sides of the equation, giving us

$$\widehat{\text{Satellites}}(Y = 1) = e^{-2.6500} * e^{0.1493(\text{Width})} * e^{-0.1997(\text{Color 3})} * e^{-0.4364(\text{Color 4})} * e^{-0.4474(\text{Color 5})}$$

or,

$$\widehat{\text{Satellites}}(Y = 1) = e^{-2.6500+0.1493(\text{Width})-0.1997(\text{Color 3})-0.4364(\text{Color 4})-0.4474(\text{Color 5})}$$

Thus for every 1 unit increase in carapace width, the number of satellites is increased by a factor of $e^{0.1493}$, or 1.161.

(C) Construct and interpret a 99% confidence interval for the beta coefficient of WIDTH in the context of this problem. (5 points)

```
round(exp(coef(model1)), 4)
```

```
## (Intercept)      WIDTH      COLOR_3      COLOR_4      COLOR_5
##      0.0706      1.1611      0.8190      0.6464      0.6393
```

```
round(exp(confint(model1, level = 0.99)), 4)
```

```
## Waiting for profiling to be done...
```

```
##              0.5 % 99.5 %
## (Intercept) 0.0156 0.3221
## WIDTH      1.1001 1.2248
## COLOR_3     0.5611 1.2420
## COLOR_4     0.4142 1.0315
## COLOR_5     0.3712 1.0967
```

```
confint(model1, "WIDTH", level = 0.99)
```

```
## Waiting for profiling to be done...
```

```
##      0.5 %      99.5 %
## 0.09536736 0.20274487
```

```
exp(0.09536736)
```

```
## [1] 1.100063
```

```
exp(0.20274487)
```

```
## [1] 1.22476
```

Since taking the natural exponent of each bound of the confidence interval returns the range [1.10, 1.22], which does not include 0, we can be 99% certain that the variable WIDTH has a positive correlation with the number of satellites a female horseshoe crab will have. This means that for every 1 unit increase in WIDTH, there we are 99% confident the number of satellites will increase by a factor lying between 1.10 and 1.22.

(D) Is WIDTH a useful predictor of satellites? Conduct an appropriate chi-square test. (8 points)

null hypothesis: WIDTH is not a statistically significant predictor of the number of satellites a female horseshoe crab will have.

```
summary(model1)
```

```
##
## Call:
## glm(formula = SATELLITES ~ WIDTH + COLOR_3 + COLOR_4 + COLOR_5,
##      family = "poisson", data = Crab)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -2.65004    0.58802  -4.507 6.58e-06 ***
## WIDTH        0.14934    0.02084   7.166 7.73e-13 ***
## COLOR_3     -0.19969    0.15364  -1.300  0.1937
## COLOR_4     -0.43636    0.17636  -2.474  0.0133 *
## COLOR_5     -0.44736    0.20912  -2.139  0.0324 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 632.79  on 172  degrees of freedom
## Residual deviance: 559.34  on 168  degrees of freedom
## AIC: 924.64
##
## Number of Fisher Scoring iterations: 6
```

$$H_0 : \beta_1 = 0$$

$$H_a : \beta_1 \neq 0$$

$$\text{T.S.: } \chi^2 = 51.351556$$

$$\text{p-value: } P(\chi^2 \geq 1.196) \approx 7.73 \times 10^{-13}$$

We can get a chi-squared value of 7.166^2 from the summary of model1 = 51.351556, and the p-value is 7.73×10^{-13} , which is very low. It is much lower than all classical α values, such as 0.1, 0.05 and 0.01, we can reject the H_0 . Thus we can confidently conclude that width is a statistically significant predictor of the number of satellites a female horseshoe crab will have at all reasonable confidence intervals.

(E) Does COLOR contribute to the prediction of satellites? Conduct an appropriate chi-square test. (8 points)

null hypothesis: COLOR is not a statistically significant predictor of the number of satellites a female horseshoe crab will have.

```
model3 <- glm(SATELLITES ~ WIDTH, family = "poisson", data = Crab)
summary(model3)
```

```
##
## Call:
## glm(formula = SATELLITES ~ WIDTH, family = "poisson", data = Crab)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -3.30476    0.54224  -6.095  1.1e-09 ***
## WIDTH        0.16405    0.01997   8.216 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##    Null deviance: 632.79  on 172  degrees of freedom
## Residual deviance: 567.88  on 171  degrees of freedom
## AIC: 927.18
##
## Number of Fisher Scoring iterations: 6
```

```
model3
```

```
##
## Call:  glm(formula = SATELLITES ~ WIDTH, family = "poisson", data = Crab)
##
## Coefficients:
## (Intercept)      WIDTH
##      -3.305      0.164
##
## Degrees of Freedom: 172 Total (i.e. Null);  171 Residual
## Null Deviance:      632.8
## Residual Deviance: 567.9    AIC: 927.2
```

```
anova(model3, model1, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: SATELLITES ~ WIDTH
## Model 2: SATELLITES ~ WIDTH + COLOR_3 + COLOR_4 + COLOR_5
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         171      567.88
## 2         168      559.34  3   8.5338 0.03618 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$$H_0 : \beta_1 = 0$$

$$H_a : \beta_1 \neq 0$$

$$\text{T.S.: } \chi^2 = 8.5338$$

$$\text{p-value: } P(\chi^2 \geq 1.196) \approx 0.03618$$

We can get a chi-squared value of 8.5338 from the Analysis of Variance test, and a p-value of 0.03618, which is low. Since it is less than the classical α value of 0.05, we can reject the H_0 at the 95% confidence interval. Thus we can conclude that color is a statistically significant predictor of the number of satellites a female horseshoe crab will have.